Independent pair-coherence readout and the native analyzer boundary

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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1 Purpose and scope

The exact question is whether one actual preparation and instrument independently identify the pair quadrature controlling exchange and predict an unused-for-fitting response with controlled uncertainty. Preserve all P1, PAH and EXCH sources, states and time laws. EXCH's finite-K side-16 torus and $Q = 11008/27$ remain their own mathematical scope; they are not identified with the native trapped gas. No damping or new model is added. IDEA006/008/011 were retrieved first. This audit uses exactly two primary sources and one Donley magnetic-pulse protocol. The verdict is `HOLD_FOR_EVIDENCE`, not a physical impossibility theorem. There is no new R or negative-result ID and no C6/T054 promotion.

The preregistered contract, fixed before verifier execution and any physical fitting, has SHA-256

`4117b174f9b8abafbee4947053a4c6f5579f994851a27d1bf0e316050cb59daf`

Its published outcomes were already seen during source intake: the reserved setting is retrospective and unused for fitting, never blind or prospective.

2 What the two actual sources provide

Donley et al., https://arxiv.org/pdf/cond-mat/0204436v2, PDF pp. 1–5, provide the native ^{85}Rb preparation, two field pulses, hold and destructive atom imaging. The readout also includes a field ramp, expansion and release. Remnant and burst atoms are separated in images; other channels are not detected. Population-time curves are fitted, rather than used as independently calibrated pre-readout pair tomography. Changing pulse2's fall time changes the measured relative phase. This establishes real phase-sensitive recombination, not its independent inverse calibration.

Goral, Kohler and Burnett, https://arxiv.org/pdf/cond-mat/0407627v2, PDF pp. 2,5–6, Eqs. 2–12, provide a native-channel functional with

$$U = U_{\#2} U_{\mathrm{evolve}} U_{\#1}, \qquad T_{2B} = D_{2B} + A_{2B}.$$

The fringe phase includes both pulses and the quasi-continuum contribution. The two-body return probability is explicitly an analogue of remnant fraction, not the calibrated destructive-imaging instrument. The source warns that a two-level mean-field description omits relevant continuum physics. Neither an ideal $\pi/2$ pulse nor the two-body analogue is substituted for the actual prepared many-body gas.

Both PDFs were hash-checked and relevant pages read/rendered by root, with independent source review. Exact pins/locators are in the contract and assessment. This bounded search does not prove global absence.

3 The independent-readout test

For a native instrument setting j, the complete calibrated effect would be $E_j = \mathcal{I}_j^*(M_j)$, including pulse2 and all later detection operations. In a closed unitary description this reduces to $U_{\#2}^\dagger M_j U_{\#2}$ only if M_j already includes the full post-pulse chain. Endpoint populations after independently calibrated analyzers can legitimately reveal coherence; population differentiation or refitting the same unknown state/analyzer phase is not independent. Changing the evolution time generally changes the state and is not tomography of a single frozen pre-readout preparation.

For clarity, the following standard finite-dimensional algebra is a *conditional diagnostic*, not a model of the native continuum. On the reference basis $|A\rangle = |2,0\rangle$, $|M\rangle = |0,1\rangle$,

$$\rho = (I + r\cdot\sigma)/2, \quad B = b^\dagger a^2 = \sqrt{2}|M\rangle\langle A|, \quad C = \mathrm{tr}(\rho B) = (r_x - i r_y)/\sqrt{2}.$$

With $H_{\mathrm{pair}} = gB + \bar{g}B^\dagger$, the reference number flux is $\dot{N}_a = -4\Im(gC)/\hbar$. Thus $\Im C$ alone is a current quadrature only after anchoring the phase convention of g. A joint basis rephase of C and g is a coordinate change, not a physical-current ambiguity. No identification with EXCH coefficients or physical time is imported.

If calibrated effects give $m_j = e_{0j} + e_j \cdot r$, write $y = Ar$ for $y = m - e_0$. A target $t = a^T r$ is identifiable exactly when

$$\ker A \subseteq \ker a^T \quad \Longleftrightarrow \quad a^T \in \mathrm{rowspan}\, A.$$

Sufficiency follows from $A^T\beta = a$. Necessity follows by two small opposite perturbations along a null direction in the state's relative interior. Known population constraints must be included in the admissible difference space. Full tomography is not necessary: a calibrated pure-y effect can suffice for the minimum target. Conversely, equal populations and calibrated x, z effects do not distinguish $(I \pm \sigma_y)/2$, whose pair quadratures and a calibrated y response differ. For a nonconvex preparation class use its actual admissible differences, not an unjustified linear enlargement. Unit rows in the primary rank controls denote centered/rescaled Pauli expectations, not raw probability effect vectors; the independent calculation uses actual projector effects.

For an analyzer with unknown orientation relative to fixed g,

$$m_\phi = \tfrac{1}{2}[1 + r_x\cos(\phi+\delta) + r_y\sin(\phi+\delta)].$$

Rotating $r_\perp$ by α and replacing δ by $\delta + \alpha$ leaves all nominal-setting probabilities unchanged while the fixed-g current can change. This diagnoses missing relative analyzer calibration, not impossibility of calibration or of the native experiment.

4 Errors and the reserved response

If $A^T\beta = a$, then $\hat{t} = \beta^T y$ and $\mathrm{Var}\,\hat{t} = \beta^T\Sigma\beta$. For a known holdout row $e_h = A^T\beta_h$,

$$\mathrm{Var}(m_h - \hat{m}_h) = \Sigma_{hh} + \beta_h^T\Sigma_{\mathrm{train}}\beta_h - 2\beta_h^T\Sigma_{\mathrm{train},h}.$$

Neither covariance diagonality nor independence is assumed. Independently justified calibration is not presumed statistically uncorrelated. It has additional joint sensitivity and higher-order remainder terms. Under independently justified effect bounds $|\Delta e_{0j}| + \|\Delta e_j\| \le \eta_j$, a fixed linear estimator has an error contribution bounded by $\sum_j |\beta_j|\eta_j$ on the Bloch ball. This is not a measured native bound. Covariance alone is not a guaranteed interval without a declared coverage rule or tail bound.

Loss outcomes and raw normalizations must be retained. Renormalizing by postselected survival generally makes the signal nonlinear in the initial state and invalidates a linear-effect rank test unless separately justified. The actual source has shot-number variation, undetected channels and state/readout

contamination. These cannot be replaced by an exact fixed-N preparation or a zero leakage/error assumption.

The published pulse2 fall-time settings 11 and 159 microseconds are frozen as a base sensitivity case and a reserved retrospective case. Their published phase fits are not independent analyzer calibration. The 159 setting has not been fitted here, but its outcome was already inspected; it receives no prospective credit. No phase prediction, holdout residual, confidence interval or empirical PASS is computed without the missing instrument/error/outcome packet. This is NOT_EXECUTED, not a failed numerical prediction.

5 Devil's-advocate review

The single missing input is a source-native, phase-referenced *analyzer-and-outcome packet*: full-chain effects on a specified pre-readout state class, anchored to the conversion coupling, with continuum/population/loss contamination, joint covariance and higher-order errors, valid finite time range and separate role-frozen outcome counts. The reviewed sources do not supply this input. The physical gate is HOLD. No existing TECT proposition is refuted by lack of this experimental packet.

1. Fringe phase is directly $\Im C$: UPHELD against that import; both pulse phases and continuum enter the actual signal.
2. Any endpoint population readout is circular: DISMISSED; independently calibrated analyzers provide valid tomography.
3. Two-level reference matrices describe the native gas: UPHELD prohibition; they are standard diagnostic controls only.
4. A convention rephase proves opposite physical current: DISMISSED; the ambiguity test holds current coupling fixed and varies analyzer/state.
5. Diagonal covariance or loss renormalization is harmless: UPHELD against it; full joint errors and survival normalization matter.
6. Published pulse variants are untouched holdout data: UPHELD against prospective credit; roles are retrospective and no fitting occurs.
7. An algebra/Lean PASS supplies missing calibration: UPHELD prohibition. No new theorem is promoted; no new Lean proof is needed for an absent physical datum. Existing Lean sources remain unchanged.

Primary and independent operator/algebra checks are separately reproducible. AI source/hostile review is not human refereeing; external review is invited. Standard inverse-problem mathematics is reused, with no new R/no-go ID.

6 Result footer

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Result ID: C6-SPACETIME-SIGNATURE / EXCH004 audit; no new R.
Precise statement: Physical readout/validation HOLD; one missing packet.
Scope: One native protocol/two sources; reference algebra conditional.
Dependencies: EXCH003/R580 boundary and source crosswalk.
Evidence grade: INHERITED, ANALYTIC, EXACT, EXECUTED, CONDITIONAL.
Reproduction command:
  python -B -X utf8 verification/scripts/exch004_verify.py
Expected output: PASS_AUDIT_COMPLETE_PHYSICAL_HOLD.
Falsification gate: New calibrated packet or exact source/math error.
Tier before / after: C6 T1 OPEN / T1 OPEN; T054 unchanged.
No-overclaim statement: No empirical PASS, native impossibility,
  quantization, physical clock, PreA, QFT, gravity or continuum.
Next required action: One phase-referenced native packet; no auto-chain.
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